

Eric Chen

(763)-310-9908 | erichchen@stanford.edu | <https://eric-h-chen.github.io/>

Education

Stanford University

Stanford, CA

B.Sc. & M.Sc. Computer Science & B.Sc. Mathematics, GPA: 4.113/4.0

Expected: June 2027

Relevant Coursework: LLMs from Scratch (CS 336), Algorithms, Graduate Probability Theory, Programming Abstractions and Data Structures (C++), Real Analysis, Deep Learning for CV (Python, PyTorch, NumPy), Computer Organization and Systems (C), Statistical ML, Stochastic Processes, NLP with DL, OS Principles (C, C++), Deep RL, Regression & ANOVA, Convex Optimization.

CS/Machine Learning Experiences

Stanford Artificial Intelligence Lab (SAIL): Chain-of-Thought and Mixture of Graders for LLM Judges in Math

Stanford, CA

First Author, Machine Learning Researcher

Oct 2025 – Present

- Designed and ran 13 judge prompting experiments across 5 LLMs using vLLM, batched inference on H200/B200 GPUs.
- Built a 500-problem human-graded benchmark with analytic rubrics; analyzed correlation and variance metrics in SciPy.
- Trained a ModernBERT grader router cutting judge MSE up to 27.8% across 4 LLM judges ($p < 10^{-4}$).
- Led a team of four (one undergrad, PhD, postdoc and faculty advisor); Published **ICLR (LLM Reasoning) & ICML (AI4MATH)**.

Stanford Artificial Intelligence Lab (SAIL) and CURIS Summer Research Internship, Stanford University

Stanford, CA

First Author, Machine Learning Researcher

Jan 2024 – July 2025

- Extracted and varied data from competitions to make Putnam-AXIOM, 522-large benchmark for SOTA LLMs' reasoning.
- Evaluated and supervised fine-tuning (FT & PEFT) SOTA LLMs (Mistral-7B, LLAMA-8B, DeepSeek) on a virtual GPU.
- Exposed contamination: fine-tuning lifted accuracy 23%→80% on originals, 12%→33% on variants.
- Spearheaded a team of peers; collaborated with Prof. Sanmi Koyejo, **published at ICML, and NeurIPS (MATH-AI track)**.

Hudson River Trading (HRT) Summer Internship

New York City, NY

Algorithm Developer (Quant Researcher) Intern

May 2026 – Aug 2026

- Trained equity return prediction models on high-frequency market data across the ingestion, features, and training/validation.
- Analyzed an alternative equity dataset and engineered features from it to improve daily return predictions.

Deep Reinforcement Learning Final Research Project, Supervised by Prof. Chelsea Finn, Stanford University

Stanford, CA

Student Machine Learning Researcher

April 2025 – June 2025

- Built AlphaZero-style RL agent for Gomoku using MCTS and self-play, boasting 100% win-or-tie rate against human players.
- Awarded **extra credit** by Prof. Finn for novel model architecture, proficiency in PyTorch, GPU optimization, and RL training.

Roblox Corporation Summer Internship

San Mateo, CA

Machine Learning & Software Engineer Intern

June 2025 – Sept 2025

- Designed & productionized transformer and set-transformer-based model to embed & aggregate collections of user data.
- Deployed new model to production at scale w/ Apache Airflow, Kafka, Hive, Spark, AWS S3, and Kubeflow infrastructure.
- Increased alt-account detection precision 2x to 92%**; enabled enforcement against high priority bad actors.

Publications, Conferences Presentations, and Professional Services

[Mixture of Graders: Adaptive Strategy Routing for LLM-Based Mathematical Evaluation](#)

(Lead Author) ICML 2026 (AI4MATH).

[Putnam-AXIOM: A Functional and Static Benchmark for Measuring Higher Level Mathematical Reasoning](#)

(First Author) ICML 2025 and NeurIPS 2024 (MATH-AI). Featured on YCombinator HackerNews.

[Rethinking LLM Judges: Chain-of-Thought and Multi-Step Pipelines for Math Grading](#)

(Lead Author) ICLR 2026 (LLM Reasoning).

[Using Machine Learning on New Feature Sets Extracted From 3D Models of Broken Animal Bones](#)

Published in the Journal of Human Evolution, Vol. 187 (Feb. 2024). (from AMAAZE, University of Minnesota)

[Unitary Conditions for Lamé and Heun Differential Operators](#)

Presented at the 2023 Joint Mathematics Meeting conference. Published in the MIT Math Department. (from MIT PRIMES-USA)

[Coarse Graining to Expedite Molecular Dynamics Simulations of Solvated Fibrinogen](#)

Presented at the Materials Research Society, Fall 2023 Meeting. (from the Garcia Research Program, Stony Brook University)

[Reviewer for the ICML AI4MATH 2025 & NeurIPS MATH-AI 2025/2026 Conference Workshops.](#)

Awards and Honors

Phi Beta Kappa, Stanford University (Beta of California)

May 2026

- Elected as a Junior, one of 36 students selected from the Class of 2027.
- The nation's oldest academic honor society; election requires distinction within the major and across a breadth of fields.

Skills and Interests

- Programming Languages:** LaTeX (mastery), Python (mastery), C++ (proficient), C (proficient), Java (intermediate).
- Frameworks and Libraries:** PyTorch, NumPy, Pandas, SciPy, SymPy, Multiprocessing, GitHub, Linux CLI.
- Languages:** English (fluent), Mandarin (fluent), Spanish (Intermediate).
- Hobbies:** piano, singing, cycling, escape rooms, competitive video games, poker, exploring the outdoors.